

TD2 : PYTHON CSV JASON

```
@SaktiJaga →/workspaces/esilv-marketpulse-g07-totally_spies (BrancheSakti) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
```

1. Which function reads JSON? → `load_instruments()`
2. Which function reads CSV? → `load_prices()`
3. Which function selects one ticker? → `filter_prices()`
4. Which function coordinates execution? → `main()`
5. Where are the input files stored? → Dans le dossier indiqué par `DATA_DIR` dans `src/main.py`

```
cat data/sample/instruments.json

[2] ✓ 0.1s

{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

```
import json
with open("data/sample/instruments.json", encoding="utf-8") as file:
    instruments = json.load(file)

[3] ✓ 0.0s

> print(type(instruments))
print(instruments.keys())

[4] ✓ 0.0s

... <class 'dict'>
dict_keys(['instrument', 'benchmark'])
```

```
import csv
def load_prices():
    with open("data/sample/prices.csv", encoding="utf-8") as file:
        return list(csv.DictReader(file))
```

✓ 0.0s

```
prices = load_prices()

print(type(prices))
print(type(prices[0]))
print(prices[0])
```

✓ 0.0s

```
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00'}
```

```
print(prices[0]["close"])
print(type(prices[0]["close"]))
```

✓ 0.0s

250.00

<class 'str'>

```
close = float(prices[0]["close"])
```

✓ 0.0s

```
def filter_prices(prices, ticker):
    ... return [row for row in prices if row["ticker"] == ticker]
```

✓ 0.0s

```
instrument_prices = filter_prices(  
    prices,  
    instrument["ticker"],  
)
```

```
benchmark_prices = filter_prices(  
    prices,  
    benchmark["ticker"],  
)
```

[17] ✓ 0.0s

```
print(len(instrument_prices))  
print(len(benchmark_prices))
```

[18] ✓ 0.0s

... 21
21

```
def get_first_close(prices):  
    first_observation = prices[0]  
    close = first_observation["close"]  
    return float(close)
```

0]

✓ 0.0s

```
~  
aapl_prices = filter_prices(prices, "AAPL")  
sp500_prices = filter_prices(prices, "SP500")  
  
print(get_first_close(aapl_prices))  
print(get_first_close(sp500_prices))
```

1]

✓ 0.0s

250.0

6600.0

```
def get_last_close(prices):  
    last_observation = prices[-1]  
    close = last_observation["close"]  
    return float(close)
```

[22] ✓ 0.0s

```
>   
apl_prices = filter_prices(prices, "AAPL")  
sp500_prices = filter_prices(prices, "SP500")  
  
print(get_last_close(apl_prices))  
print(get_last_close(sp500_prices))
```

[23] ✓ 0.0s

... 266.2
6742.0

```
def display_market_summary(asset, prices, show_currency=True):
    ticker = asset["ticker"]
    name = asset["name"]
    observations = len(prices)
    first_close = get_first_close(prices)
    last_close = get_last_close(prices)

    print(f"{ticker} - {name}")
    print(f"Observations : {observations}")

    if show_currency:
        currency = asset["currency"]
        print(f"First close : {first_close:.2f} {currency}")
        print(f>Last close : {last_close:.2f} {currency}")
    else:
        print(f"First close : {first_close:.2f}")
        print(f>Last close : {last_close:.2f}")

    print()
```

✓ 0.0s

```
display_market_summary(instrument, instrument_prices, show_currency=True)
display_market_summary(benchmark, benchmark_prices, show_currency=False)
```

✓ 0.0s

```
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
```